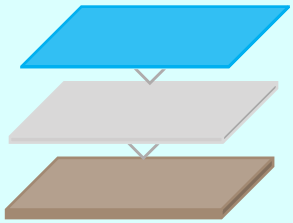
Composite materials

Factsheet

Composites are created when 2+ materials are bonded together → design it to come apart



Graphic layer
Printed SAV or direct UV print onto ACP/ substrate

→ **If choosing SAV – make it removable**

ACP sheet
Sheeting for facades/ hoardings/ covers

→ **Choose AUS standard ACP (see below) + design to separate**

Plywood or timber frame
Typical substrate

→ **Choose ply which can be recycled**

Its all in the glue! How you bond layers determines the waste fate
Use debonding methods and mechanical fixtures (Button-fix, screws, rails or heat-debondable tape)

Choose the graphic method before choosing the panel

OPTION	BEST FOR	TRADE-OFFS
Printed SAV bonded to a substrate or ACP	Short-life campaigns, frequent rebranding	Allows graphic replacement without replacing the panel; may have adhesive residue risk and end-of-life separation issues
Direct UV print to ACP	Large-format branding/ one-off event graphics	Fast, no vinyl waste, sharp finish; harder to change graphics later
Graphic system - fabric, mesh, PVC banners	Reusable overlay assets and circular event systems	Highest reuse potential and recovery

Go-to graphics options

PRODUCT	SUPPLIER	PRICE	CIRCULAR (RECYCLABLE/ REUSABLE)
Fabric façades	Tetrik Structures	Low cost	Sublimation printing. Recovery solutions in progress
Removeable SAV (PVC-free)	Visionary Digital, Avery 2900/ 2611 / 3111 / 3041 / 2002 3M graphic film AAC large format	Price competitive	Yes - can be recycled
PVC banners (sail track)	Spicers, Shann, Halifax Vogel, Ricky Richards	Standard market rates	Yes - can be recycled via TA take-back program
Weathertex cladding	Weathertex	Price competitive	Yes - can be recycled
Water-based paint (Wattyl ID, FX, Dulux Enviro)	Wattyl, FX Australia, Dulux	Standard market rates	Yes - can be recycled. Choose waterbased for timber use so that it can be recycled
Paintback tinted recycled paint	Paintback	30% cheaper than commercial products	Yes – and cheapest in market. Australian made 100% circular with take-back program

ACP – see below

Choose Australian Standard Non-Combustible Aluminium Composite Panel (ACP)

PRODUCT	KEY FIRE / COMPLIANCE CREDENTIALS	CIRCULAR (RECYCLABLE/ REUSABLE)
Fairview Vitracore G2 (bonded aluminium panel)	• Tested to AS 1530.1-1994 – laminate layers "NOT deemed combustible" • AS 1530.3: Ignitability 0, Spread-of-Flame 0, Heat Evolved 0 • Used where NCC requires non-combustible external walls	Yes – reusable + recyclable when separated from substrate
PROBOND PROCORE A1 (profiled aluminium core panel)	• Marketed as DtS non-combustible to AS 1530.1 • Also tested to AS 1530.3 • Zero polyethylene; aluminium core structure	Yes – reusable + recyclable when separated from substrate
Alcadex A1 (mineral-matrix core ACP)	• CSIRO-tested to AS 1530.1-1994 – "NOT deemed COMBUSTIBLE" • Mineral matrix core (no PE) between aluminium skins	Yes – reusable + recyclable when separated from substrate
Fairview Vitradual (3 mm solid aluminium cassette system)	• Deemed non-combustible to AS 1530.1 • Solid aluminium; no polymer core	Yes – reusable + recyclable when separated from substrate

Before you order ACP ensure you comply with TA Fire/Safety

Only ACP tested to **AS 1530.1-1994** and **"NOT deemed combustible"** is accepted. Treat any "non-combustible ACP" claim as unverified without test report and will be **rejected**.

Ask the supplier for

- The full AS 1530.1-1994 test report from an accredited lab (e.g. CSIRO, Warringtonfire), dated on or after 1 July 2017.
- What was tested: the whole panel, or laminate layers only.
- A conclusion that states "NOT deemed combustible".
- Confirmation the panel you order matches the tested one: core, skin thickness, adhesive, coating.

Be wary of

- "FR", "A2" or "90% mineral core" claims without an AS 1530.1 report. Many still fail and are deemed combustible.
- Avoid polyethylene core
- Reports to other standards only (EN 13501-1 A2, BS 8414, AS 5113). Useful for system performance, but not proof of non-combustibility under the NCC.